

Shyam Sundar Singh

AI / ML Engineer — Machine Learning • Generative AI
• Computer Vision

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summary

Data Science / ML Engineer with 2+ years of production experience designing, training, and deploying ML models, including applied work in deep learning, computer vision, and generative AI (CLIP, HuggingFace Transformers, meta-learning, few-shot/zero-shot classification). Skilled across the full ML lifecycle — data preprocessing, model development in TensorFlow/PyTorch, and deployment of production-ready Flask REST APIs on AWS. Seeking an AI/ML Engineer role building GenAI-enabled, cloud-deployed applications.

experience

Data Science Engineer

Feb 2024 — Present

VIVAX.IN

- Designed and deployed classification and regression models for customer behavior analysis, increasing prediction accuracy by 18%.
- Deployed predictive ML models via production Flask REST APIs serving 500+ daily requests, handling the full pipeline from training to live inference.
- Built a customer churn prediction model (Random Forest) achieving 82% accuracy, contributing to a 12% improvement in customer retention.
- Standardized data cleaning and transformation pipelines across recurring projects, cutting feature noise by 35% and improving model reliability.
- Automated daily reporting workflows with Python scripts, reducing manual processing time by 70% (~10 hrs/week).

Software Development Intern

Sep 2022 — Nov 2022

NBLIK

- Redesigned an analytics dashboard UI, increasing average user session duration by 40%.
- Built an equipment monitoring module for production processes, reducing unplanned downtime by 20%.

relevant projects

AI Trust Level Prediction System

Apr 2026

Python • Scikit-learn • XGBoost • LightGBM • CatBoost • Pandas • NumPy

Built a predictive analytics pipeline estimating AI Trust Level (%) from AI perception datasets, with an ensemble of CatBoost, XGBoost, and LightGBM validated via K-Fold CV and RandomizedSearchCV. Conducted data preprocessing, feature engineering, and missing value treatment to improve model performance.

Low-Resource Image and Text Classification using Meta-Learning

Mar 2026

Python • PyTorch • HuggingFace Transformers • CLIP • Scikit-learn

Built a meta-learning pipeline (Prototypical Networks, MAML) reaching 80%+ 5-way accuracy with 5 labeled samples/class; applied CLIP for zero-shot recognition of unseen categories from text alone, cutting annotation needs by 90%. Benchmarked zero-shot vs. few-shot vs. supervised approaches across datasets with fewer than 50 samples.

Real-Time Face Detection and Counting System

May 2024

Python • YOLOv4 • OpenCV

Fine-tuned YOLOv4 on 5,000+ annotated images, achieving 92% mAP at 30+ FPS for real-time detection.

Personalized Loan Demand Prediction — MachineHack Hackathon

Jul 2025

Python • XGBoost • LightGBM • CatBoost • Scikit-learn

Cleaned 10,000+ customer records (95% missing values); tuned 4 ensemble models across 100+ parameter combinations, ranking top 15% among 500+ participants.

technical skills

ML & Deep Learning

TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, LightGBM, CatBoost, neural network design & tuning

GenAI / NLP / LLMs

HuggingFace Transformers, CLIP, prompt engineering, meta-learning (Prototypical Networks, MAML), zero/few-shot inference

Computer Vision

OpenCV, YOLOv4, object detection, image processing, real-time video pipelines

MLOps / Deployment

Flask REST APIs in production, AWS, Git/GitHub, Postman, data pipeline standardization

Data Science

Predictive modeling, feature engineering, Pandas, NumPy, SQL, MySQL, MongoDB, Tableau, Matplotlib, Seaborn, Plotly

Languages

Python, SQL, Java, JavaScript

education

B.Tech, Computer Science & Engineering

Amity University, Lucknow Campus

May 2024

certifications & achievements

Git and GitHub

Udemy — Certificate of Completion

MachineHack Hackathon

Ranked top 15% among 500+ participants

Jan 2025

Internship Certificate

NBLIK, via Internshala

Nov 2022